

Develop a program that executes the equivalent of the shell command `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code:

```
geya@LenovoL480: ~/OS/Pra x + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int pipefd[2];

    pipe(pipefd);

    // Create first child for ls
    if (fork() == 0) {

        close(pipefd[0]);

        // Send stdout of ls into pipe
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", "-l", NULL);

        return 0;
    }

    // Create second child for grep
    if (fork() == 0) {
```

```

        close(pipefd[1]);

        // Take input of grep from pipe
        dup2(pipefd[0], STDIN_FILENO);

        close(pipefd[0]);

        execlp("grep", "grep", ".c", NULL);

        return 0;
    }

    // Parent closes pipe
    close(pipefd[0]);
    close(pipefd[1]);

    // Wait for both children
    wait(NULL);
    wait(NULL);

    return 0;
}
|

```

Output:

```

geya@LenovoL480:~/OS/Practical/Week 5$ gcc task2.c -o task2
geya@LenovoL480:~/OS/Practical/Week 5$ ./task2
-rw-r--r-- 1 geya geya 1231 Sep 14 20:29 task1.c
-rw-r--r-- 1 geya geya 785 Sep 14 20:33 task2.c
geya@LenovoL480:~/OS/Practical/Week 5$ |

```